NurtureNet: Explainable Five-Modality Video Safety Assessment for Children and Adolescents Using Gated Cross-Attention

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***Abstract*—Children increasingly consume short-form videos in which harmful content may be conveyed through visual scenes, speech, background audio, on-screen text, or contextual metadata. This paper presents NurtureNet, a five-modality video-safety assessment system that integrates video, acoustic audio, automatic speech recognition (ASR), optical character recognition (OCR), and platform metadata. Modality representations are fused using a Gated Cross-Attention Network (GCAN) with explicit availability masks for handling missing inputs. The model predicts Safe, Harmful Content, Adult Content, or Suicide, after which a transparent age-aware policy maps probabilities to Allow, Caution, or Block decisions for ages 7–12 and 13–17. On a cleaned corpus of 4,676 videos, GCAN achieves 80.06% accuracy and 79.93% Macro-F1, compared with 71.51% accuracy and 71.53% Macro-F1 for a controlled video-only baseline. Ablation experiments further demonstrate the contribution of individual modalities and the model's behavior under missing inputs.**

***Keywords*—child content safety, multimodal video classification, gated cross-attention, missing modalities, content moderation, age-aware moderation**

I. Introduction

A. Background and Importance

Short-form video distributes meaning across visual scenes, speech, environmental sound, captions, overlays, and platform context. Consequently, visually benign content may contain harmful dialogue or audio, while on-screen text may alter the meaning of otherwise ambiguous scenes. Recent child-safety research therefore increasingly considers audiovisual and multimodal evidence rather than vision alone [1]–[3].

Practical systems must also handle incomplete inputs because speech, readable OCR, metadata, or even individual media streams may be unavailable. NurtureNet addresses these issues by separating *content-risk prediction* from *age-policy action*. A learned classifier predicts Safe, Harmful Content, Adult Content, or Suicide, while a transparent policy converts the probability vector into Allow, Caution, or Block for ages 7–12 and 13–17.

B. Problem Context

Existing approaches cover visual video classification [4], [5], audiovisual child-safety assessment [1], [3], subtitle and metadata analysis [6], and multimodal hate detection [2]. However, these systems generally employ different subsets of available evidence and different safety taxonomies. NurtureNet investigates whether five separately extracted modalities can be jointly fused while explicitly accounting for unavailable inputs.

C. Problem Statement

The problem is to combine visual, acoustic, spoken, written, and contextual evidence to classify video risk while remaining operational when selected modalities are unavailable. The study additionally examines whether multimodal GCAN improves balanced classification performance over a controlled video-only baseline and how performance changes when modalities are removed.

D. Aim and Objectives

The aim is to develop and evaluate a multimodal video-safety assessment pipeline for children and adolescents. The objectives are to: (i) construct a deduplicated and leakage-safe dataset; (ii) extract five aligned modality representations; (iii) develop availability-aware GCAN fusion; (iv) compare it with a video-only baseline; (v) evaluate missing-modality behavior; and (vi) produce inspectable predictions with age-aware decisions.

E. Research Questions

RQ1: Does five-modality GCAN outperform a VideoMAE-feature baseline?

RQ2: How does removal of individual or combined modalities affect GCAN?

RQ3: How do restricted modality combinations affect performance?

RQ4: Can inference expose probabilities, modality availability, gate weights, and textual evidence while applying explicit age-aware rules?

F. Contributions

The main contributions are: (i) a five-modality GCAN with availability-aware attention and gating; (ii) a cleaned 4,676-video corpus with leakage-safe splits; (iii) controlled video-only and missing-modality evaluations; and (iv) integration of four-class prediction with transparent age-aware policy decisions and inspectable evidence.

II. Related Work

A. Existing Approaches

Visual safety classifiers demonstrate strong image and video representations but cannot directly capture speech or

acoustic evidence [4], [5], [7], [8]. Audiovisual systems address this limitation by combining visual and acoustic information. Ahmed *et al.* incorporate non-speech audio for children's-video moderation [1], while SNIFR aligns audio and visual streams using cross-transformers [3].

Textual information provides additional context. Samba combines subtitles with metadata for young-child video suitability [6], while hateful-meme research demonstrates that visual and textual evidence can interact in ways not observable from either modality independently [9]–[11]. MM-HSD extends multimodal moderation to video using frames, audio, transcripts, and OCR [2]. Age-adaptive approaches further motivate differentiated decisions for different age groups [12].

Table I summarizes representative recent studies in reverse chronological order. Their scores are not directly comparable because datasets, taxonomies, and evaluation protocols differ.

B. Critical Analysis

The literature demonstrates that complementary modalities can improve safety assessment, but modality coverage varies considerably. Visual methods omit speech and audio, audiovisual methods generally omit OCR and metadata, and richer multimodal systems often address different safety taxonomies. Moreover, missing-modality robustness and explanation faithfulness remain limited areas of evaluation. NurtureNet therefore combines five channels with explicit availability masks while treating gate weights as inspectable model signals rather than causal explanations.

C. Research Gap

The reviewed studies reveal four main gaps. First, visual-only systems cannot capture harmful speech, acoustic cues, or textual overlays. Second, audiovisual approaches generally do not combine OCR and contextual metadata with visual, acoustic, and spoken evidence. Third, richer multimodal moderation studies often target hate speech or other taxonomies rather than age-aware child-content safety. Finally, robustness to unavailable modalities and transparent separation between learned risk prediction and age-specific policy remain underexplored. NurtureNet addresses these gaps through five-modality fusion, explicit availability masks, learned gating, controlled modality-removal analysis, and a transparent age-policy layer.

III. Methodology and Experimental Setup

A. Proposed Methodology

As shown in Fig. 1, NurtureNet contains five stages: data preparation, modality extraction, availability-aware fusion, risk classification, and age-aware policy mapping. For video x, representations e_v, e_a, e_s, e_o, and e_m correspond to video, acoustic audio, ASR, OCR, and metadata. An availability vector $m \in \{0,1\}^5$ identifies usable modalities.

VideoMAE represents 16 uniformly sampled video frames. BEATs encodes general acoustic information from 16-kHz mono audio. Whisper large-v3-turbo generates speech transcripts, which are encoded using multilingual BERT (mBERT). PP-OCRv5 extracts on-screen text from sampled frames and mBERT encodes the resulting text. Metadata combines a 768-dimensional mBERT representation of title and description with

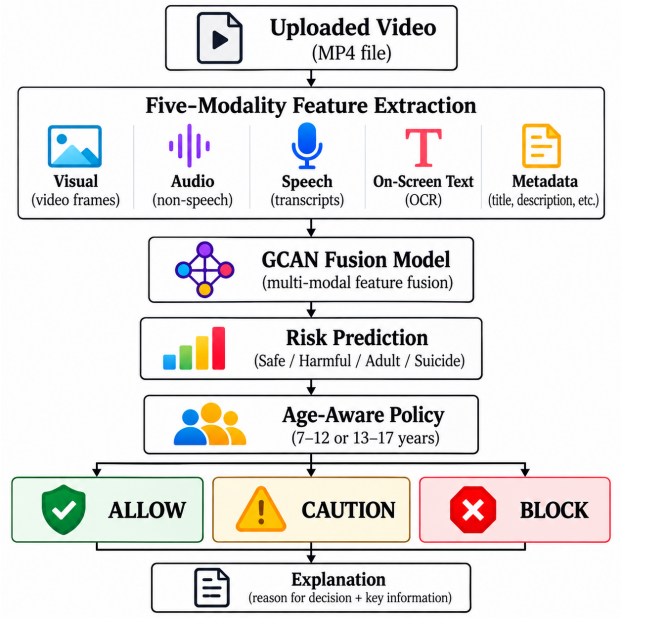


Fig. 1. NurtureNet end-to-end multimodal pipeline with five modality extractors, availability-aware GCAN fusion, four-class prediction, and age-policy routing.

21 structured features, producing a 789-dimensional vector. The remaining modalities are 768-dimensional.

GCAN predicts four-class logits followed by softmax probabilities. A separate deterministic policy then maps these probabilities to age-specific actions.

B. Model Architecture

Each modality is projected into a shared $d = 256$ space:

$$h_i = \text{Dropout}\left(\text{GELU}\left(\text{LN}(W_i e_i + b_i)\right)\right). \tag{1}$$

The five projected representations form modality tokens $H \in \mathbb{R}^{5\times 256}$. Eight-head cross-attention contextualizes these tokens, while unavailable modalities are excluded using a key-padding mask.

A learned gate produces one logit for each contextualized modality. After masking unavailable modalities, normalized weights are

$$\alpha_i = \frac{\exp(g_i)}{\sum_{j:m_j=1}\exp(g_j)}, \qquad \alpha_i = 0 \text{ if } m_i = 0. \tag{2}$$

The fused representation is

$$h_f = \sum_{i=1}^{5} \alpha_i h_i. \tag{3}$$

The classifier maps $256 \to 128 \to 4$. The final GCAN contains 1,586,053 parameters, uses dropout 0.35, and applies modality dropout of 0.10 during training.

TABLE I. LITERATURE SURVEY

Ref.	Year	Method	Dataset	Metrics	Performance	Key Limitation
[13]	2026	Multimodal harmful-video analysis	186,727 videos	Acc., P, R, F1	94% multiclass Acc.	Platform-specific taxonomy
[8]	2025	CNN/ViT models including DenseNet201	CVID (664 samples)	Acc., P, R, F1	∼99% Acc.	Small image-only dataset
[2]	2025	Cross-modal frames, audio, ASR, OCR	HateMM	Macro-F1, Acc.	87.4% F1; 88.1% Acc.	Hate-speech rather than child safety
[3]	2025	Cascaded audio–visual cross-transformer	Child harmful-content benchmark	Acc., Macro-F1	89.6% Acc.; 88.2% F1	Audio–visual only
[7]	2024	Transfer learning with classical classifiers	Social-media images (∼3k)	Acc., P, R, F1	82% Acc.	Image-only; small dataset
[1]	2024	Audiovisual children's-video moderation	Children's-video benchmark	Acc., P, R, F1	88.7% Acc.; 87.2% F1	No OCR, ASR, or metadata
[4]	2024	Spatio-temporal video transformers	TikHarm	Acc., P, R, F1	Transformer-based results	Primarily visual evidence
[14]	2024	Multimodal harm taxonomy and annotation	19,422 videos	Acc., P, R, F1	57.9% Macro-F1	Different deployment objective
[12]	2024	Age estimation + content classification	YouTube dataset	Acc., P, R, F1	96.89% Acc.	Additional privacy assumptions
[15]	2023	Explainable recommendation dataset and baselines	REASONER	F1, NDCG, RMSE, MAE	41.68% F1	Recommendation rather than safety
[11]	2023	Contextual explanation generation	3,228 hateful memes	BLEU, ROUGE, BERTScore	0.473 BERTScore	Static hate-specific data
[9]	2022	VinVL + OCR with Random Forest	Facebook Hateful Memes	Acc., AUROC	68.4% Acc.; 76.8% AUROC	Static binary hate task
[10]	2022	PromptHate with RoBERTa	FHM and HarM	AUC, Acc.	81.45 / 90.96 AUC	Hate-specific task
[6]	2022	BERT subtitles + GRU metadata	70K YouTube videos	Accuracy	95%	Depends on subtitle/metadata availability
[5]	2022	EfficientNet-B7 + BiLSTM	111,156 cartoon clips	Acc., F1	95.66% Acc.	Visual-focused pipeline

TABLE II. FINAL DATASET DISTRIBUTION

Split	Safe	Harm.	Adult	Suic.	Total
Train	868	828	783	794	3273
Validation	186	177	168	170	701
Test	186	178	167	171	702
Total	1240	1183	1118	1135	4676

C. Age-Aware Policy Layer

For ages 7–12, Adult, Suicide, or Harmful probability ≥ 0.35 produces Block; otherwise unsafe probability $1 - p(\text{Safe}) \geq 0.40$ produces Caution, while Safe confidence ≥ 0.60 produces Allow. Remaining cases receive Caution.

For ages 13–17, Adult or Suicide probability ≥ 0.50 produces Block; Harmful probability ≥ 0.50 or aggregate unsafe probability ≥ 0.45 produces Caution; Safe confidence ≥ 0.55 produces Allow. These thresholds are manually defined policy parameters rather than learned or clinically validated developmental criteria.

D. Dataset and Preprocessing

The study uses TikHarm and TikHarm Extended, two public TikTok harmful-content datasets obtained from Kaggle [16], [17]. The datasets contain short-form TikTok videos with safety-related class labels used in this study as Safe, Harmful Content, Adult Content, and Suicide.

The original TikHarm dataset contains 3,948 records and the extended dataset contains 775 records, giving 4,723 records before cleaning. The datasets were first combined and compared using video IDs. Conflicting duplicate IDs were identified, including eight IDs with inconsistent labels, and duplicate records were resolved during cleaning. This process produced a final set of 4,676 unique videos.

Stratified sampling produced 3,273 training, 701 validation, and 702 testing videos, with zero video-ID leakage between splits.

Video, audio, ASR, and OCR feature arrays have shape 4676×768, metadata has shape 4676×789, and modality masks have shape 4676×5. Of the 4,676 samples, 4,207 contain all five active modalities and 467 have inactive OCR. One sample has video and OCR inactive, while another has audio and ASR inactive. These cases are explicitly represented by modality masks.

E. Training Configuration

GCAN V4 uses hidden dimension 256, eight attention heads, dropout 0.35, modality dropout 0.10, and four output classes. Weighted cross-entropy uses class weights $[1.00, 1.15, 1.00, 1.00]$ for Safe, Harmful Content, Adult Content, and Suicide. AdamW is used with learning rate 10^{-4}, weight decay 10^{-4}, batch size 64, and gradient clipping at 1.0. Training is capped at 40 epochs with early-stopping patience 8 and a ReduceLROnPlateau scheduler.

Five configurations were evaluated. V3 obtained the highest validation Macro-F1 (0.8190), while V4 obtained 0.8165 but increased Harmful recall from 0.6949 to 0.7232. V4 was therefore selected before test evaluation. Its checkpoint corresponds to epoch 12. Experiments used Python 3.11.6, PyTorch 2.11.0+cu128, CUDA 12.8, and an NVIDIA RTX A4000.

F. Evaluation Metrics

Performance is evaluated using accuracy, precision, recall, F1, Macro-F1, cross-entropy loss, and the confusion matrix. For class c,

$$P_c = \frac{TP_c}{TP_c + FP_c}, \qquad R_c = \frac{TP_c}{TP_c + FN_c}, \tag{4}$$

$$F1_c = \frac{2P_cR_c}{P_c + R_c}. \tag{5}$$

Macro-F1 is the primary model-selection metric. Final results are point estimates from one held-out test split.

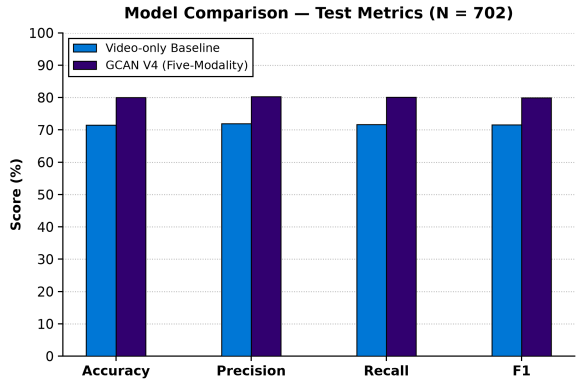


Fig. 2. Test performance of the video-only baseline and five-modality GCAN V4 ($N = 702$).

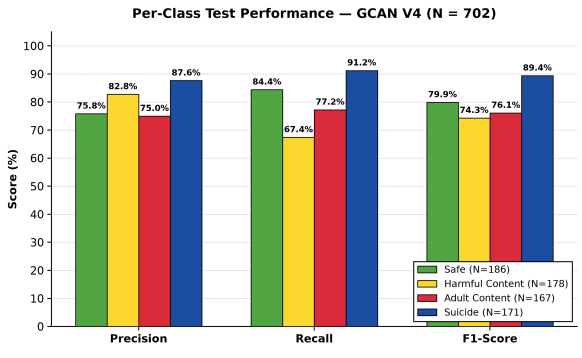


Fig. 3. Per-class precision, recall, and F1-score of GCAN V4.

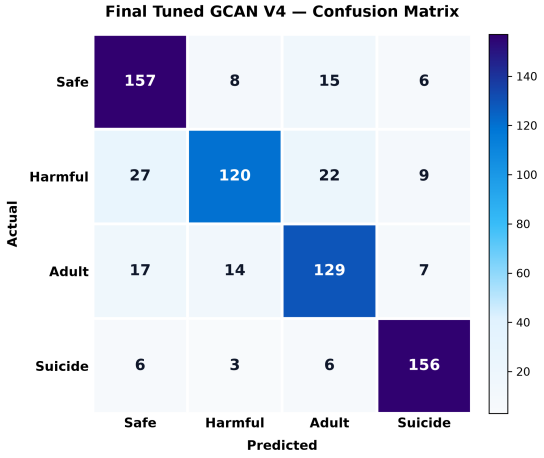


Fig. 4. Confusion matrix of GCAN V4 on the 702-video test set.

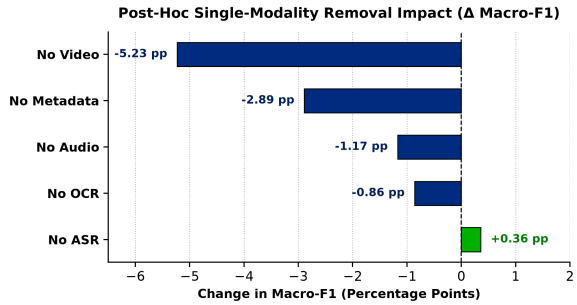


Fig. 5. Macro-F1 change after post-hoc removal of individual modalities.

IV. Experiments and Results

A. Experimental Design

Four experiments were performed: evaluation of GCAN V4 on the 702-video test set, comparison with a separately trained VideoMAE-feature MLP, post-hoc removal of individual and combined modalities, and qualitative inspection of correctly classified test samples. Modality-removal experiments modify availability masks without retraining and are therefore interpreted as diagnostic ablations.

B. Quantitative Results

GCAN V4 achieved 80.06% test accuracy and 79.93% Macro-F1, compared with 71.51% accuracy and 71.53% Macro-F1 for the controlled video-only baseline. Multimodal fusion therefore improves accuracy by 8.55 and Macro-F1 by 8.40 percentage points.

Table III consolidates the principal quantitative findings, including the controlled baseline, single-modality removals, combined removals, and restricted-modality inference. Dashes indicate that only Macro-F1 was recorded for the corresponding diagnostic configuration.

Per-class performance is shown in Fig. 3. Suicide achieves the highest F1 (89.40%) and recall (91.23%), while Harmful Content is the most difficult class with 74.30% F1 and 67.42% recall.

The confusion matrix in Fig. 4 shows that Harmful Content is most frequently confused with Safe (27 cases) and Adult Content (22 cases), while 156 of 171 Suicide samples are correctly classified.

1) Missing-Modality Analysis: Removing video causes the largest Macro-F1 decrease (5.23 percentage points), followed by metadata (2.89), audio (1.17), and OCR (0.86). Removing ASR produces a small 0.36-point increase, suggesting possible redundancy or noise in this specific test condition rather than general ASR ineffectiveness.

Combined removal further reduces performance: video+audio gives 67.82% Macro-F1, video+metadata 70.18%, and audio+ASR+OCR 73.66%. Restricted inference gives 63.60% for video only, 61.79% for audio only, 43.56% for ASR only, 46.33% for OCR only, and 59.25% for metadata only. The complete five-modality configuration achieves the strongest balanced performance among the evaluated settings.

C. Training Analysis

V3 achieved validation Macro-F1 0.8190, whereas V4 achieved 0.8165 but improved Harmful recall from 0.6949 to 0.7232 using a 1.15 class weight. Increasing the weight to 1.25 in V5 reduced Macro-F1 to 0.8128. V4 was therefore retained as an explicit safety-oriented trade-off; test results were not used for this selection.

TABLE III. COMPREHENSIVE QUANTITATIVE RESULTS ON THE HELD-OUT TEST SET ($N = 702$)

Category	Configuration	Accuracy (%)	Macro-F1 (%)	Δ Macro-F1 (pp)
Main model	Five-modality GCAN V4	80.06	79.93	0.00
Baseline	VideoMAE-feature MLP	71.51	71.53	−8.40
Single removal	Without video	–	74.70	−5.23
Single removal	Without metadata	–	77.04	−2.89
Single removal	Without audio	–	78.76	−1.17
Single removal	Without OCR	–	79.07	−0.86
Single removal	Without ASR	–	80.29	+0.36
Combined removal	Without video + audio	–	67.82	−12.11
Combined removal	Without video + metadata	–	70.18	−9.75
Combined removal	Without audio + ASR + OCR	–	73.66	−6.27
Restricted inference	Video only	–	63.60	−16.33
Restricted inference	Audio only	–	61.79	−18.14
Restricted inference	ASR only	–	43.56	−36.37
Restricted inference	OCR only	–	46.33	−33.60
Restricted inference	Metadata only	–	59.25	−20.68

D. Qualitative Results

Table IV presents the highest-confidence correct prediction from each class. The Harmful example has unavailable OCR, for which both the availability mask and gate weight are zero, demonstrating the implemented masking behavior.

Gate weights vary between examples, but they are interpreted only as internal model weighting signals rather than causal explanations.

E. Computational Performance

The GCAN fusion/classification stage was benchmarked using precomputed features for all 702 test samples. Across 100 synchronized runs, mean batch time was 1.6535 ms and peak allocated GPU memory was 52.98 MB. These measurements exclude modality extraction and therefore do not represent end-to-end video inference.

F. Results Summary

GCAN improves Macro-F1 over the controlled video-only baseline by 8.40 percentage points. Video has the largest single-modality removal impact, followed by metadata. The model remains operational when selected modalities are masked, and qualitative results confirm zero gating for unavailable evidence.

V. DISCUSSION

A. Interpretation of Findings

The improvement from 71.53% to 79.93% Macro-F1 indicates that complementary modalities provide information not captured by the visual baseline alone, consistent with prior audiovisual moderation research [1], [3]. However, the experiment evaluates the complete GCAN configuration and does not isolate whether the gain originates specifically from gating, attention, or increased model capacity.

Ablation results identify video as the dominant modality, while metadata also provides substantial complementary information. Audio and OCR contribute smaller aggregate improvements. Harmful Content remains the most difficult class, with 67.42% recall, whereas Suicide achieves 91.23% recall. This suggests greater semantic heterogeneity within the Harmful category.

B. Comparison with Existing Literature

Existing research differs from NurtureNet in both modality coverage and task definition. TikGuard primarily analyzes visual TikTok evidence [4]; Ahmed *et al.* and SNIFR use audiovisual information but do not jointly include ASR, OCR, and metadata [1], [3]; MM-HSD uses frames, audio, ASR, and OCR but targets hate-speech detection rather than child-content safety [2]; and age-adaptive work introduces age-sensitive decisions but under a different pipeline and dataset [12]. In contrast, NurtureNet jointly evaluates five modalities, explicitly masks unavailable inputs, studies single and combined modality removal, and separates four-class risk prediction from age-aware policy decisions.

Because these studies use different datasets, class definitions, and evaluation protocols, their reported scores are not treated as direct head-to-head numerical comparisons. The controlled comparison within this study shows GCAN reaching 79.93% Macro-F1 compared with 71.53% for the video-only baseline, an improvement of 8.40 percentage points.

C. Limitations

The dataset is primarily TikTok-derived, so cross-platform generalization has not been established. Evaluation uses a single train/validation/test split without repeated-seed confidence intervals or significance tests. Post-hoc modality ablations alter masks without retraining specialized subset models. Harmful Content recall remains comparatively low, and OCR/ASR errors may propagate into text representations. Age-policy thresholds are manually specified rather than expert validated, and gate weights have not been validated as causal explanations. Finally, the reported runtime measures only GCAN fusion with precomputed features rather than complete MP4 processing.

D. Future Work

Future work should evaluate larger multi-platform and creator-disjoint datasets, repeated-seed experiments, probability calibration, and improved Harmful Content classification. GCAN should also be compared with matched-capacity concatenation, late fusion, and ungated attention. Explanation faithfulness requires perturbation-based and human evaluation, while age-policy thresholds should be validated with relevant

TABLE IV. VERIFIED QUALITATIVE EXAMPLES FROM GCAN V4

True Class	Video ID	Confidence	Class Probabilities [S,H,A,Su]	Availability Mask	Gate Weights
Safe	7282361849658592513	0.975839	[0.9758, 0.0104, 0.0094, 0.0044]	[1,1,1,1,1]	[0.2287,0.2278,0.2276,0.1194,0.1966]
Harmful Content	6928836923398049030	0.987355	[0.0058, 0.9874, 0.0036, 0.0032]	[1,1,1,0,1]	[0.2412,0.3504,0.0919,0.0000,0.3165]
Adult Content	7221906826533539098	0.983817	[0.0063, 0.0078, 0.9838, 0.0021]	[1,1,1,1,1]	[0.2223,0.3434,0.1172,0.1296,0.1875]
Suicide	7062539048228949295	0.991265	[0.0026, 0.0038, 0.0023, 0.9913]	[1,1,1,1,1]	[0.1624,0.2417,0.1281,0.2567,0.2112]

TABLE V. GCAN V4 FUSION-STAGE BENCHMARK

Hardware	NVIDIA RTX A4000
Parameters	1,586,053
Batch size	702
Repeated runs	100
Mean batch time	1.6535 ms
Std. batch time	0.3653 ms
Latency/sample	0.002355 ms
Batched throughput	424,554.75 samples/s
Peak allocated memory	52.98 MB
Peak reserved memory	78.00 MB

experts and caregivers. End-to-end inference latency should additionally be measured from raw video input through all modality extractors.

VI. CONCLUSION

This paper presented NurtureNet, a five-modality video-safety assessment system integrating VideoMAE visual features, BEATs audio representations, Whisper/mBERT speech features, PP-OCRv5/mBERT on-screen text features, and metadata through availability-aware GCAN fusion. On a cleaned 4,676-video dataset, GCAN V4 achieved 80.06% accuracy and 79.93% Macro-F1, outperforming a controlled video-only baseline by 8.55 and 8.40 percentage points, respectively. Ablation experiments identified video as the most influential modality while demonstrating operation under missing inputs. A separate deterministic policy converts predicted probabilities into age-aware Allow, Caution, or Block decisions. Future evaluation should address cross-platform generalization, calibration, explanation faithfulness, expert-validated age policies, and end-to-end deployment performance.

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